Metastatic Melanoma in the Eye and Orbit

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Objective: Presentation of a large series of patients with metastatic melanoma involving the eye and orbit.

Design: Retrospective clinical study.

Participants: Thirteen cases of metastatic melanoma involving intraocular tissues, 6 cases of metastatic melanoma in the orbit, and 1 case of metastatic melanoma involving both the eye and the orbit, treated in Lausanne between 1986 and 2002, were identified from the computer files of the ocular oncology and orbitopalpebral surgery units of Jules Gonin Hospital.

Methods: The analysis is based on the demographic data concerning these patients, clinical data concerning the tumor, the treatment applied, the outcome, and the follow-up.

Main Outcome Measures: Clinical presentation of intraocular and orbital metastasis, interval between primary tumor and ocular metastasis, survival of the patients, evaluation of various therapeutic protocols.

Results: Intraocular metastases (14 cases, 15 eyes) were situated in the choroid in 11 cases (isolated lesion, 6 cases; multiple lesions, 3 cases; diffuse involvement, 2 cases), in the iris and ciliary body in 2 cases, and in the retina and vitreous in 2 other cases. The primary tumor was a cutaneous melanoma in 8 cases, a melanoma of the contralateral eye in 3 cases, a mucosal melanoma in 1 case, and was unknown in 2 cases. The mean interval between the diagnosis of ocular metastases and the patient's death was 8.8 months (range, 1–48 months). The primary tumor in the 7 cases of orbital metastases was a cutaneous melanoma in 5 cases, a uveal melanoma in the contralateral eye in 1 case, and was unknown in 1 case. The mean interval between the diagnosis of orbital metastases and death was 19.7 months (range, 5–48 months). The patients were treated by various protocols. The best results, in terms of both local tumor control and preservation of visual function, were obtained with circumscribed proton beam radiotherapy or external beam irradiation, depending on the site and extent of the tumor.

Conclusions: Metastatic melanomas to the eye and orbit are rare and generally occur in patients with disseminated metastases during the terminal stages of the disease, with a short life expectancy. Treatment is palliative and, among the various possible treatment options, circumscribed proton beam radiotherapy or global photon beam radiotherapy, at relatively high irradiation doses, seems to achieve the most favorable results. *Ophthalmology 2003;110:2245–2256 © 2003 by the American Academy of Ophthalmology.*

Melanomas are tumors that rarely metastasize to the eye and the orbit.[1] A limited number of isolated case reports or series of cases[2–9] of metastatic melanomas in the eye and orbit have been published to date. Two reviews of the literature, one performed by Gündüz et al in 1985 on intraocular metastatic melanoma and the other by Orcutt and Char in 1986 on orbital metastatic melanoma estimated the number of published cases to be 67 and 29, respectively.

The site of the primary tumor can be a cutaneous melanoma, a uveal melanoma in the contralateral eye, a mucosal melanoma, or an unknown primary tumor. Intraocular metastases tend to involve the choroid and also the anterior uvea, particularly the iris, optic disk, and, surprisingly, the retina and vitreous, with a higher frequency than metastatic carcinomas.[3–5,9,10] Periocular metastases preferentially involve the orbit. In rare cases, metastases may involve both the eye and the adnexa.[4,11–13]

The cancer-related mortality of patients with orbital and ocular metastatic melanoma is high, regardless of the site of the primary tumor, and most patients die from melanoma with a mean or median survival from the time of diagnosis of ocular metastases ranging from 2.5 to 9 months.[3,6,9,10,14]

This study, based on retrospective analysis of 13 cases of metastatic melanoma of the eye, 6 cases of metastatic melanomas of the orbit, and 1 case of metastatic melanoma involving both the eye and the orbit,[15] was designed to provide more detailed information about the clinical presentation of metastatic tumors, patient survival, and treatment of the tumor by radiotherapy.

Patients and Methods

Review of the computer files of the ocular oncology unit and the orbitopalpebral surgery unit identified 20 cases of metastatic mel-

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